LLM Mathematical Proof Audit Report

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October 21, 2026

Problem 1: Calculus – Integration Verification

AI Output Evaluation: Failed (Step 2 Logical Error)

Problem Statement

Evaluate the definite integral:

$$\int_0^\infty xe^{-x}\,dx$$

Step-by-Step AI Audit

- **Step 1:** AI set up Integration by Parts correctly ($u = x$, $dv = e^{-x}dx$).
- **Step 2 (Error):** AI evaluated boundary condition incorrectly as:

$$\lim_{x\to\infty}(-xe^{-x}) = \infty$$

- **Correction:** Applied L'Hopital's Rule to show:

$$\lim_{x\to\infty}\frac{-x}{e^x} = 0$$

Corrected Final Result

$$\int_0^\infty xe^{-x}\,dx = 1 \tag{1}$$